MAT2040 TUTORIAL 14

Yuqing Zhou

CUHK(SZ)

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Question 1

For an $n \times n$ matrix A, let λ be an eigenvalue of A. $f(x) = a_m x^m + a_{m-1} x^{m-1} + \cdots + a_1 x + a_0$ is a polynomial in x, proof $f(\lambda)$ is an eigenvalue of $f(A)$.

Solution

Let x be an eigenvector belonging to λ.

$$Ax = \lambda x, A^2x = \lambda^2 x, \cdots, A^m x = \lambda^m x.$$

Then

$$\begin{aligned} f(A)x &= (a_m A^m + a_{m-1}A^{m-1} + \cdots + a_1 A + a_0 I)x \\ &= a_m A^m x + a_{m-1}A^{m-1}x + \cdots + a_1 Ax + a_0 Ix \\ &= a_m \lambda^m x + a_{m-1}\lambda^{m-1}x + \cdots + a_1 \lambda x + a_0 x \\ &= (a_m \lambda^m + a_{m-1}\lambda^{m-1} + \cdots + a_1 \lambda + a_0)x \\ &= f(\lambda)x \end{aligned}$$

Therefore, $f(\lambda)$ is an eigenvalue of $f(A)$.

Question 2

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 2 & 3 & -2 \\ -1 & -2 & 0 \end{bmatrix}$$

Find an orthogonal matrix U that diagonalizes A.

Solution

$$p_A(\lambda) = \det(A - \lambda I_3) = \det\left(\begin{bmatrix} -\lambda & 2 & -1 \\ 2 & 3-\lambda & -2 \\ -1 & -2 & -\lambda \end{bmatrix}\right)$$

$$= -\lambda \det\left(\begin{bmatrix} 3-\lambda & -2 \\ -2 & -\lambda \end{bmatrix}\right) - 2\det\left(\begin{bmatrix} 2 & -1 \\ -2 & -\lambda \end{bmatrix}\right) - \det\left(\begin{bmatrix} 2 & -1 \\ 3-\lambda & -2 \end{bmatrix}\right)$$

$$= -\lambda(-\lambda(3-\lambda) - 4) - 2(-2\lambda - 2) - (-4 + 3 - \lambda)$$

$$= -\lambda^3 + 3\lambda^2 + 9\lambda + 5 = -(\lambda^2 + 2\lambda + 1)\lambda + 5\lambda^2 + 10\lambda + 5$$

$$= -(\lambda+1)^2\lambda + 5(\lambda+1)^2 = (\lambda+1)^2(5-\lambda)$$

Eigenvalues are $\lambda_1 = \lambda_2 = -1, \lambda_3 = 5$.
The eigenspace for $\lambda = -1$ is
$\textbf{Span}\left\{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}\right\}$. The eigenvectors $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ are not orthogonal.

Using Gram-Schmidt process, one has: $\mathbf{v}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$$\mathbf{u}_2' = \mathbf{u}_2 - < \mathbf{u}_2, \mathbf{v}_1 > \mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \frac{\mathbf{u'}_2}{\|\mathbf{u'}_2\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$\mathbf{v}_1, \mathbf{v}_2$ are orthonormal eigenvectors w.r.t. $\lambda = -1$.

The eigenspace for $\lambda = 5$ is $\mathbf{Span}\left\{\begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}\right\}$. $\mathbf{u}_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$ is the eigenvector w.r.t $\lambda = 5$.
The normalization of this vector is

$$\mathbf{v}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}.$$

$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ is an orthonormal set in $\mathbb{R}^3$. Let

$U = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{-1}{\sqrt{3}} & \frac{-2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$ be the orthogonal matrix. Then

$$U^{-1}AU = U^TAU = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Question 3

Given quadratic form $f(x_1, x_2, x_3)$
$= 5x_1^2 + x_2^2 + 5x_3^2 + 4x_1x_2 - 8x_1x_3 - 4x_2x_3$ determine it is positive definite or not.

Solution

$$A = \begin{bmatrix} 5 & 2 & -4 \\ 2 & 1 & -2 \\ -4 & -2 & 5 \end{bmatrix}, \; |A - \lambda I| = (1-\lambda)(-\lambda^2 + 10\lambda - 1), \text{ thus } \lambda_1 = 1,$$

$\lambda_2 = 5 - 2\sqrt{6}$, $\lambda_3 = 5 + 2\sqrt{6}$, it is positive definite.